

11	A	Using $\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$, $c_{\text{r.m.s.}} \propto \sqrt{T}$, $\frac{c'_{\text{r.m.s.}}}{c_{\text{r.m.s.}}} = \sqrt{\frac{160 + 273}{80 + 273}} \quad (\text{In MCQ, use 273 is faster for calculation})$ $c'_{\text{r.m.s.}} = \sqrt{\frac{160 + 273}{80 + 273}} \times 350$ $= 387 \approx 390 \text{ m s}^{-1}$

No.	Answer	Solution
12	C	Initial total kinetic energy of the gas molecules